

“Calculus 3”

Multi-Variable Calculus

Instructor: Álvaro Lozano-Robledo

Day 10



Any Reminders? Any Questions?

- I will have regular office hours 2/19 – 3:30-4:30
 - I will have additional office hours 2/19 – 4:30-5:30
 - Calc 3 Calc Night: MONT 104 at 6:30-8:30pm on Thursdays!
-
- Exam I is on Friday, Feb 20th
 - Exam practice questions/exam and solutions on HuskyCT

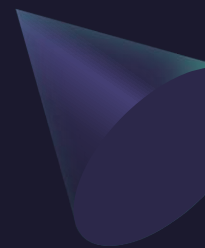
EXAM 1 -- Friday, February 20th

Exam Covers:

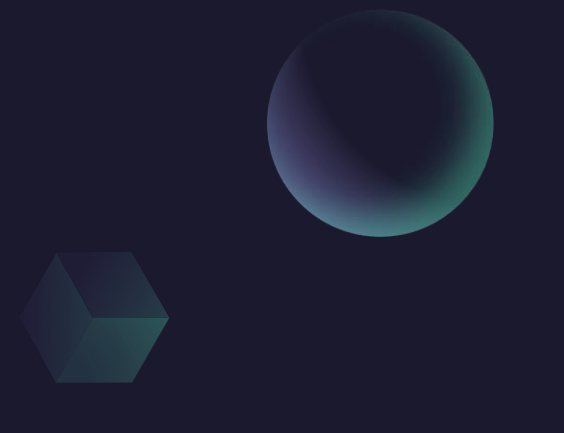
- **Chapter 12**
 - Sections 12.1 – 12.6
- **Chapter 14**
 - Sections 14.1, 14.3 – 14.8



ALVARO: Start the recording!



“Calculus 3”

A sphere, a cube, and a cone are positioned in the upper right area of the slide. The sphere is at the top right, the cube is below it, and the cone is further down and to the left.

Multi-Variable Calculus

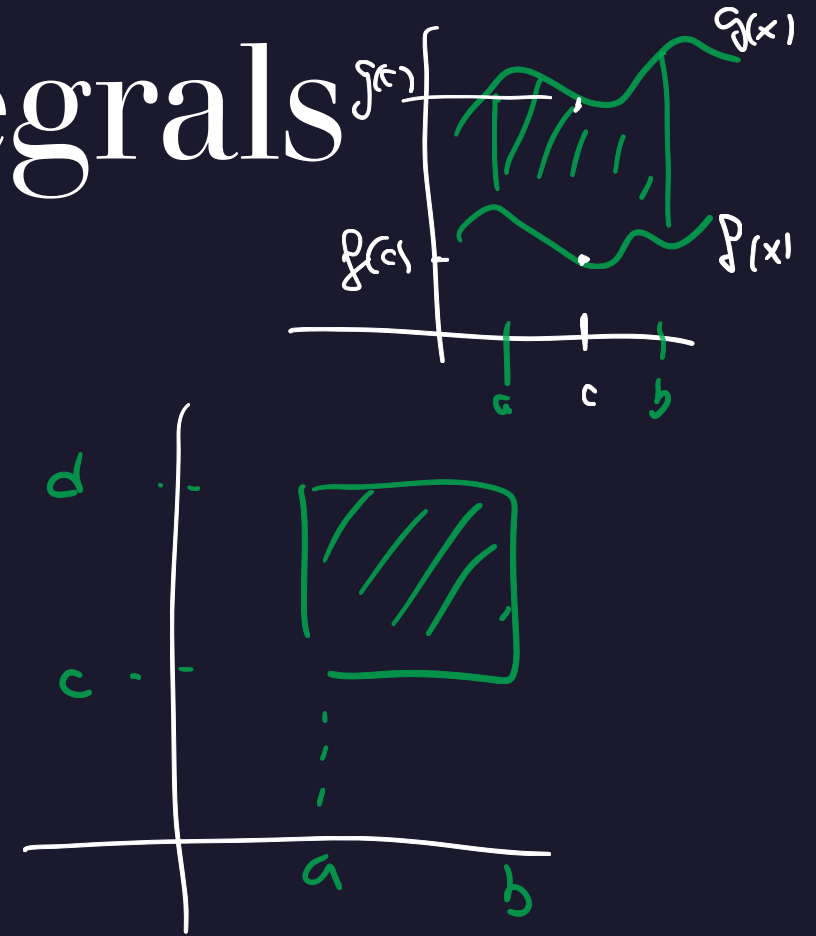
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Double Integrals over Regions

A cone and a torus are positioned in the lower right area of the slide. The cone is in the middle right, and the torus is on the far right, partially cut off by the edge of the slide.

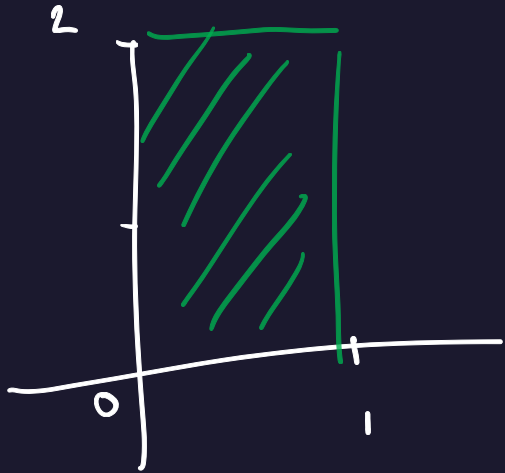
Today – Double Integrals in Regions!

- General Regions
- Regions of Type I and II
- Changing the Order of Integration
- Properties of Double Integrals



Example: (Warm up) Calculate the following iterated integral

$$\int_0^1 \left(\int_0^2 (2xy + 2y + 1) dy \right) dx$$



$$\int_0^2 2xy + 2y + 1 dy$$

$$= xy^2 + y^2 + y \Big|_0^2 = 4x + 4 + 2 - 0 - 0 - 0 = 4x + 6$$

$$\int_0^1 4x + 6 dx = 2x^2 + 6x \Big|_0^1 = 8 - 0 = \boxed{8}$$

Regions of Type I and II

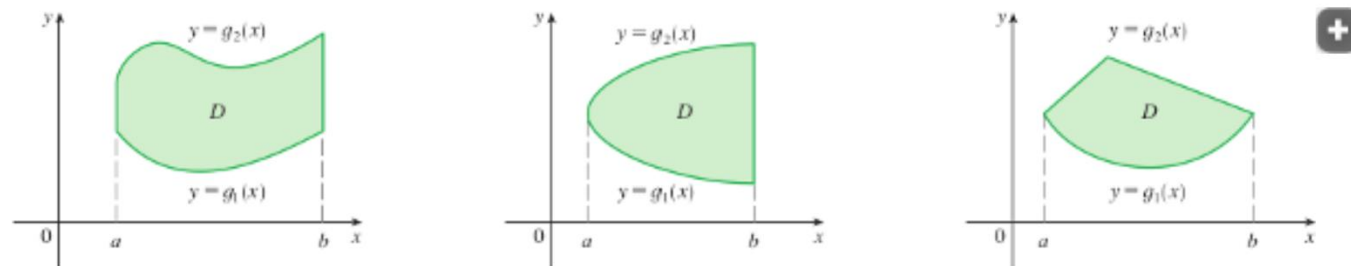
A plane region D is said to be of **type I** if it lies between the graphs of two continuous functions of x , that is,

$$D = \{(x, y) \mid a \leq x \leq b, g_1(x) \leq y \leq g_2(x)\}$$

where g_1 and g_2 are continuous on $[a, b]$. Some examples of type I regions are shown in [Figure 5](#).

Figure 5

Some type I regions



Regions of Type I and II

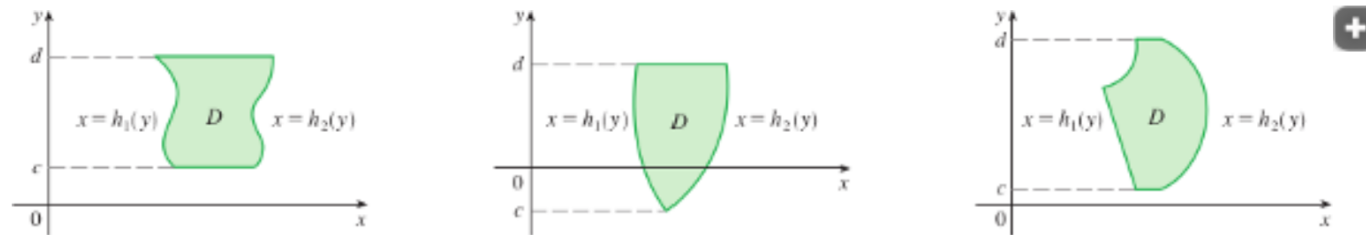
We also consider plane regions of **type II**, which can be expressed as

$$D = \{(x, y) \mid c \leq y \leq d, h_1(y) \leq x \leq h_2(y)\}$$

where h_1 and h_2 are continuous. Three such regions are illustrated in [Figure 7](#).

Figure 7

Some type II regions



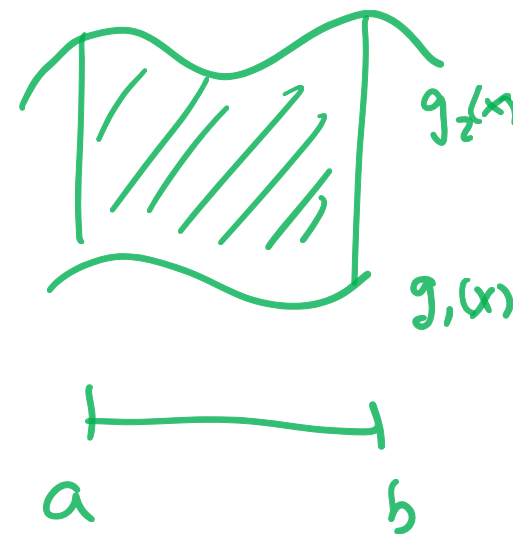
Integrals over Regions of Type I

3 If f is continuous on a type I region D described by

$$D = \{(x, y) \mid a \leq x \leq b, g_1(x) \leq y \leq g_2(x)\}$$

then

$$\iint_D f(x, y) dA = \int_a^b \int_{g_1(x)}^{g_2(x)} f(x, y) dy dx$$



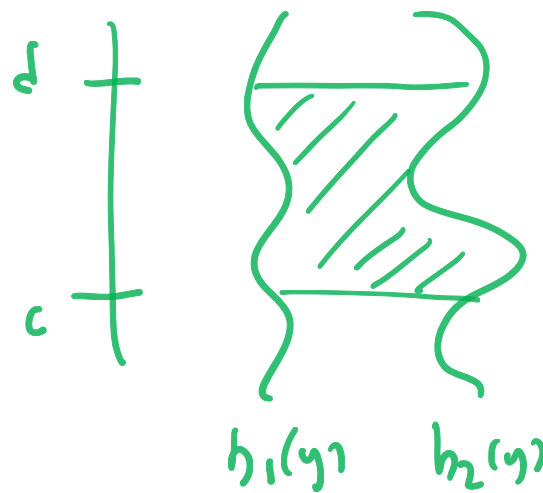
Integrals over Regions of Type II

4 If f is continuous on a type II region D described by

$$D = \{(x, y) \mid c \leq y \leq d, h_1(y) \leq x \leq h_2(y)\}$$

then

$$\iint_D f(x, y) \, dA = \int_c^d \int_{h_1(y)}^{h_2(y)} f(x, y) \, dx \, dy$$



Example: Evaluate the double integral

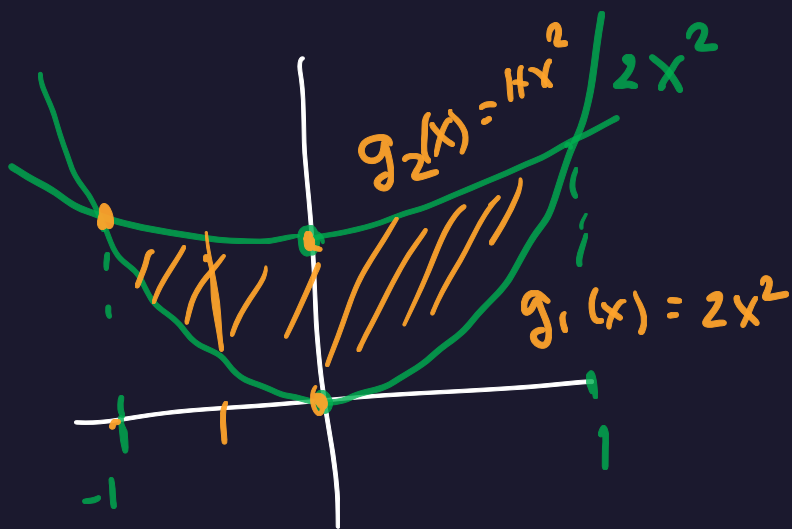
$$\iint_R (x + 2y) dA$$

where R is the region bounded by the parabolas

$$y = 2x^2 \text{ and } y = 1 + x^2.$$

$$2x^2 = 1 + x^2 \Rightarrow x^2 = 1 \Rightarrow \boxed{x = \pm 1}$$

$$R = \{ (x, y) : -1 \leq x \leq 1, 2x^2 \leq y \leq 1 + x^2 \}$$



TYPE I

$$\int_{-1}^1 \int_{2x^2}^{1+x^2} (x + 2y) dy dx$$

Example: Evaluate the double integral

$$\iint_R (x + 2y) \, dA$$

where R is the region bounded by the parabolas
 $y = 2x^2$ and $y = 1 + x^2$.

$$\int_{-1}^1 \int_{2x^2}^{1+x^2} (x + 2y) \, dy \, dx = \int_{-1}^1 \left. xy + y^2 \right|_{2x^2}^{1+x^2} dx$$

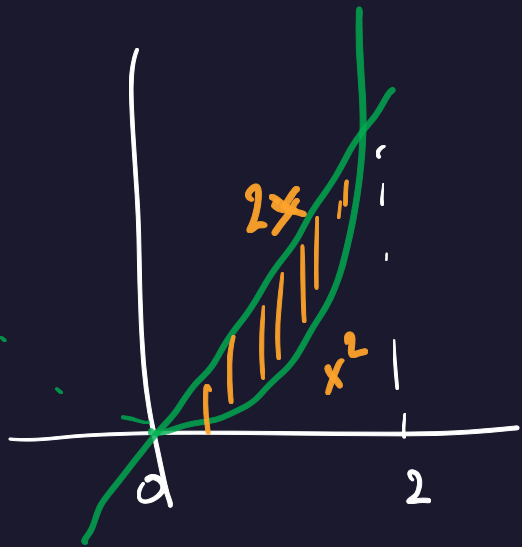
$$= \int_{-1}^1 \left(x(1+x^2) + (1+x^2)^2 - \left(x \cdot (2x^2) + (2x^2)^2 \right) \right) dx$$

$$= \int_{-1}^1 (x + x^3 + 1 + x^4 + 2x^2 - 2x^3 - 2x^4) dx$$

$$= \int_{-1}^1 (1 + x + 2x^2 - x^3 - x^4) dx =$$

$$A = \frac{32}{15}$$

Example: Find the volume of the solid that lies under the paraboloid $z = x^2 + y^2$ and above the region in the xy -plane bounded by the line $y = 2x$ and the parabola $y = x^2$. (As a Type I integral.)



$$2x = x^2 \rightarrow 2x - x^2 = 0$$

$$x(2-x) = 0 \Rightarrow \begin{matrix} x=0 \\ x=2 \end{matrix}$$

$$\int_0^2 \int_{x^2}^{2x} (x^2 + y^2) dy dx$$

Example: Find the volume of the solid that lies under the paraboloid $z = x^2 + y^2$ and above the region in the xy -plane bounded by the line $y = 2x$ and the parabola $y = x^2$.

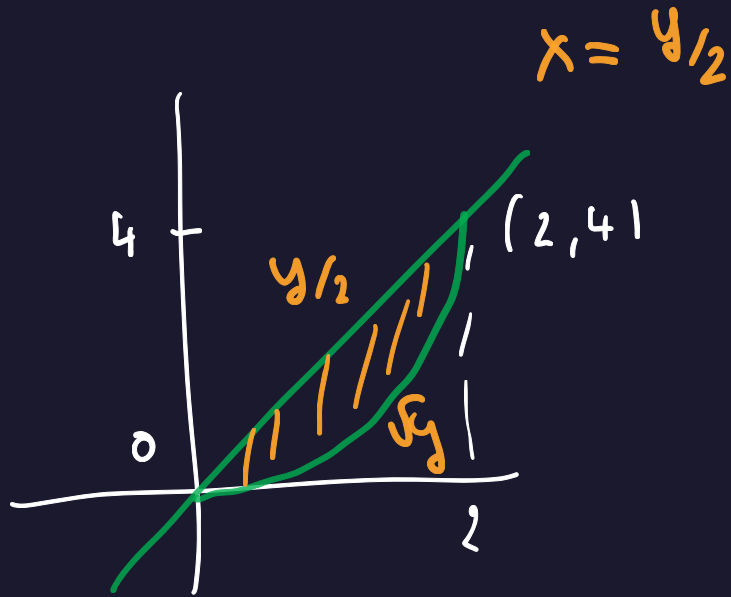
$$\int_0^2 \int_{x^2}^{2x} (x^2 + y^2) dy dx = \int_0^2 \left. x^2 y + \frac{y^3}{3} \right|_{x^2}^{2x} dx$$

$$= \int_0^2 \left(x^2 \cdot 2x + \frac{(2x)^3}{3} - x^2 \cdot x^2 - \frac{(x^2)^3}{3} \right) dx$$

$$= \int_0^2 \left(2x^3 + \frac{8x^3}{3} - x^4 - \frac{x^6}{3} \right) dx \dots$$

$$\boxed{A = \frac{216}{35}}$$

Example: Find the volume of the solid that lies under the paraboloid $z = x^2 + y^2$ and above the region in the xy -plane bounded by the line $y = 2x$ and the parabola $y = x^2$. (As a Type II integral.)

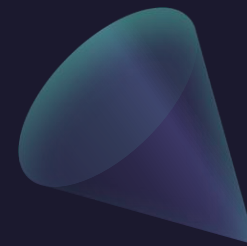
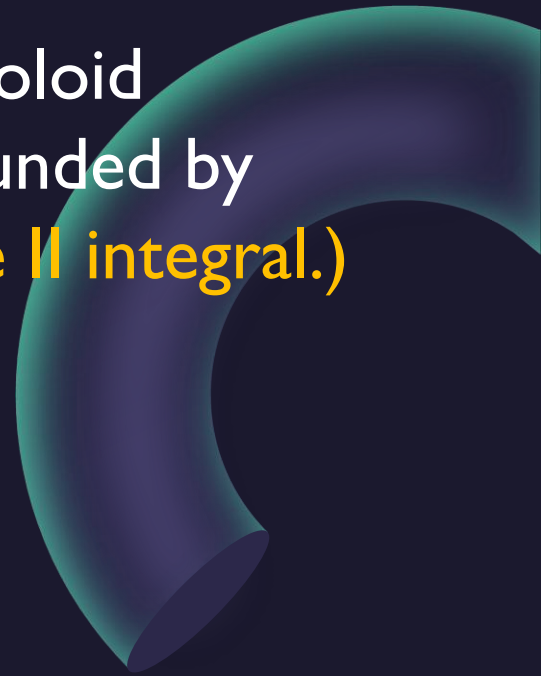


$$x = y/2$$

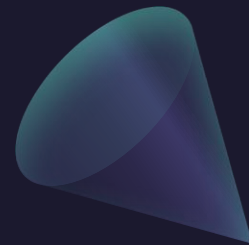
$$x = \sqrt{y}$$

$$\int_0^4 \int_{y/2}^{\sqrt{y}} (x^2 + y^2) dx dy$$

$$= \int_0^4 \left[\frac{x^3}{3} + y^2 x \right]_{y/2}^{\sqrt{y}} dy = \dots$$

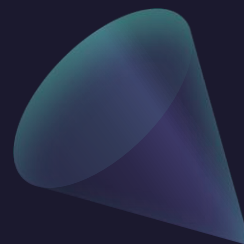
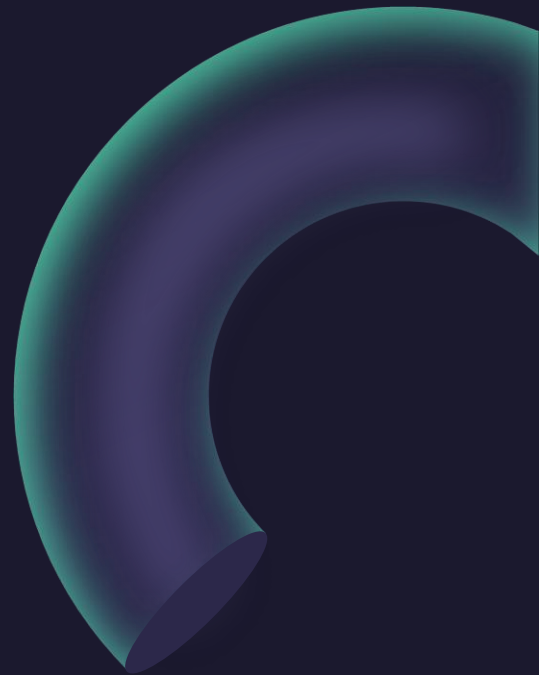


Example: Find the volume of the solid that lies under the paraboloid $z = x^2 + y^2$ and above the region in the xy -plane bounded by the line $y = 2x$ and the parabola $y = x^2$.



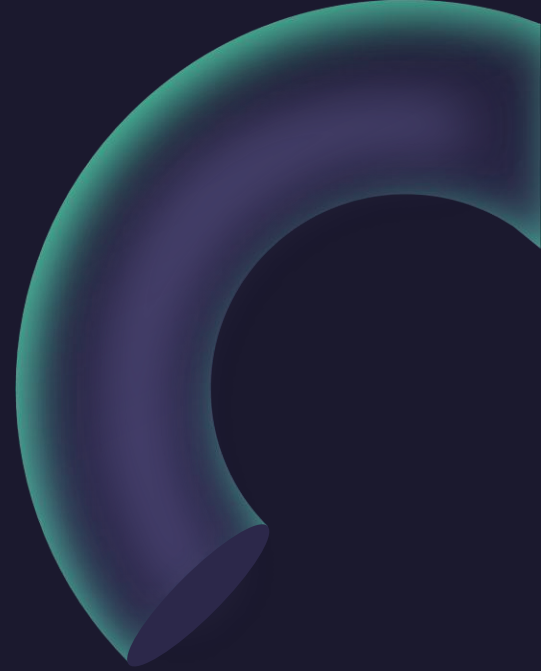
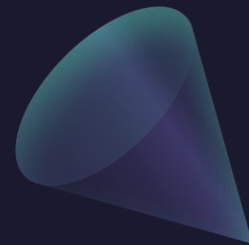
Example: Evaluate the iterated integral

$$\int_0^1 \int_x^1 \sin(y^2) \, dy \, dx$$



Example: Evaluate the iterated integral

$$\int_0^1 \int_x^1 \sin(y^2) \, dy \, dx$$



Properties of Double Integrals

$$\iint_D [f(x, y) + g(x, y)] dA = \iint_D f(x, y) dA + \iint_D g(x, y) dA$$

$$\iint_D c f(x, y) dA = c \iint_D f(x, y) dA \quad \text{where } c \text{ is a constant}$$

If $f(x, y) \geq g(x, y)$ for all (x, y) in D , then

7

$$\iint_D f(x, y) dA \geq \iint_D g(x, y) dA$$

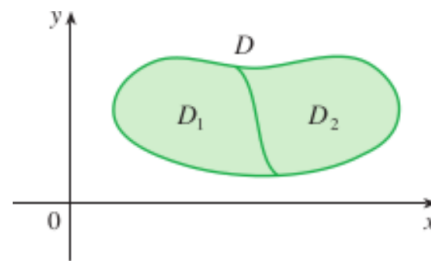
Properties of Double Integrals

If $D = D_1 \cup D_2$, where D_1 and D_2 don't overlap except perhaps on their boundaries (see [Figure 17](#)), then

8

$$\iint_D f(x, y) \, dA = \iint_{D_1} f(x, y) \, dA + \iint_{D_2} f(x, y) \, dA$$

Figure 17



Properties of Double Integrals

$$\iint_D 1 \, dA = A(D)$$

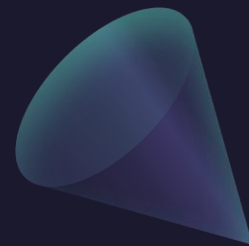
10 If $m \leq f(x, y) \leq M$ for all (x, y) in D , then

$$m \cdot A(D) \leq \iint_D f(x, y) \, dA \leq M \cdot A(D)$$

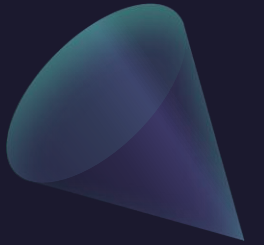
Example: Estimate the value of the double integral

$$\iint_R e^{-(x^2+y^2)} dA$$

where $R = \{(x, y): x^2 + y^2 \leq 1\}$ is the circle of radius 1.

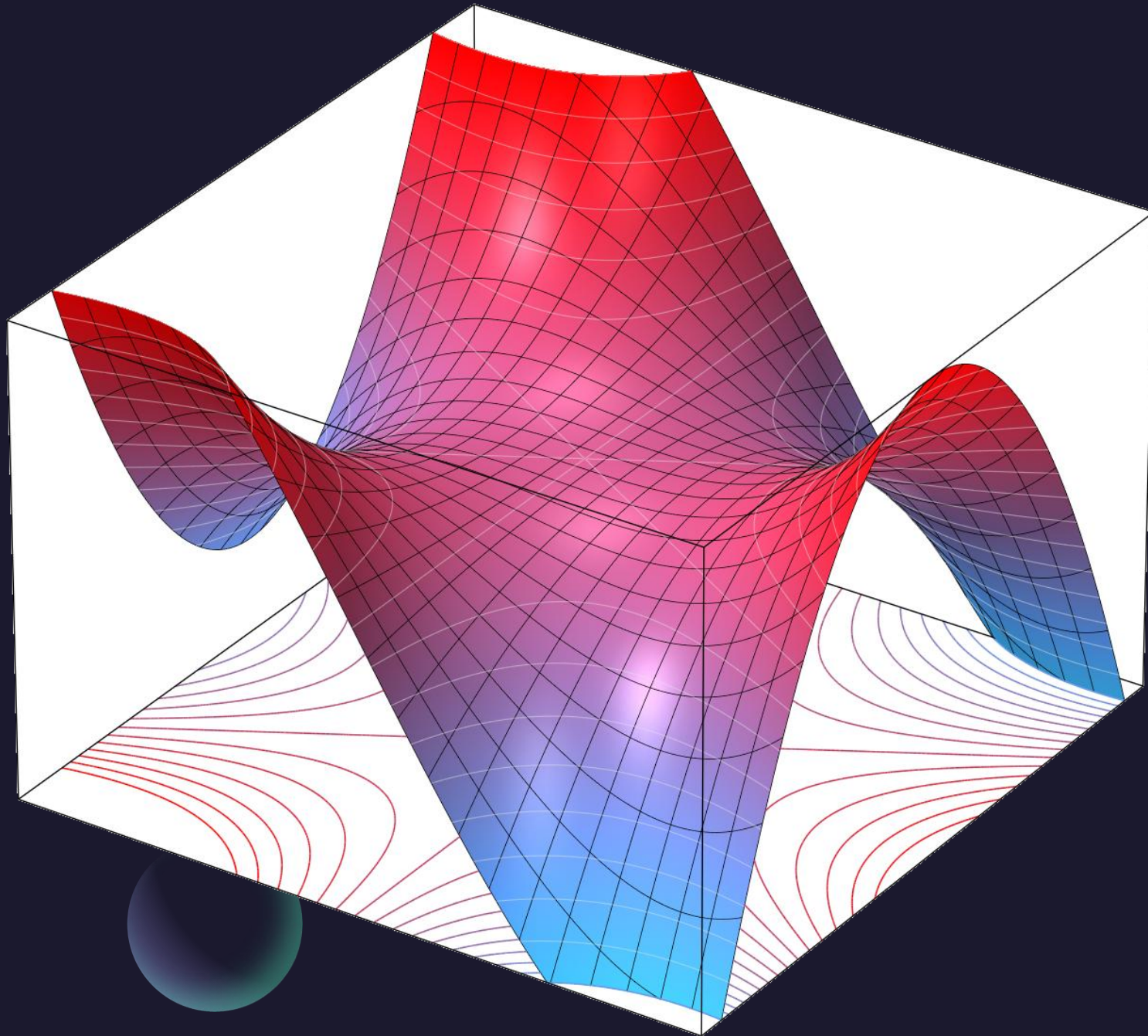


Questions?



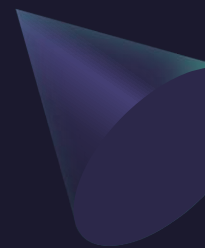
Thank you

Until next time.





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Exam I : Review



1. Let $\vec{a} = \langle 1, 1, 4 \rangle$ and $\vec{b} = \langle c, 3, 4 \rangle$, where c is an unknown constant.

(a) Find the value of c so that $\vec{a}$ and $\vec{b}$ are orthogonal.

(b) With the value of c from part (a), find $\vec{a} \times \vec{b}$.

$$a \cdot b = 0 \iff a \perp b \quad |\vec{a} \times \vec{b}|$$

$$(1, 1, 4) \cdot (c, 3, 4) = c + 3 + 16 = 0$$

$$c + 19 = 0 \rightarrow \boxed{c = -19}$$

$$a \times b = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 1 & 1 & 4 \\ -19 & 3 & 4 \end{vmatrix}$$

$$|\vec{a} \times \vec{b}| = |a||b|\sin\theta$$

1. Let $\vec{a} = \langle 1, 1, 4 \rangle$ and $\vec{b} = \langle c, 3, 4 \rangle$, where c is an unknown constant.

- (a) Find the value of c so that $\vec{a}$ and $\vec{b}$ are orthogonal.
- (b) With the value of c from part (a), find $\vec{a} \times \vec{b}$.

2. Find the equation of a line that passes through $(1, 2, 3)$ and is perpendicular to the plane $x - y + 3z = 5$.

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3. Find the equation of a plane through the origin, $(0, 1, 2)$ and $(3, 0, 1)$.

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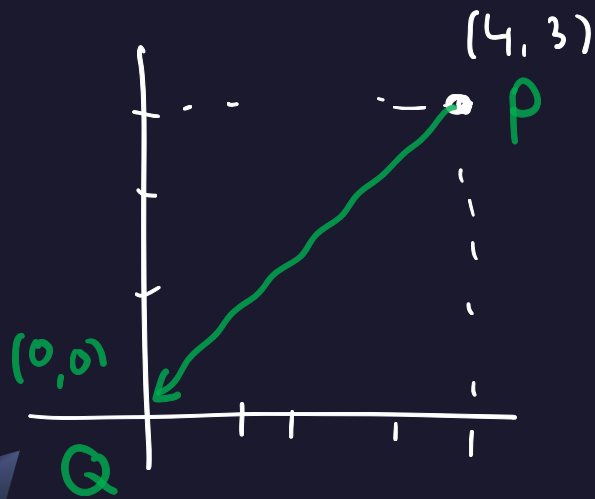
4. Let $f(x, y)$ be a function satisfying $f(4, 3) = 5$ and $\nabla f(4, 3) = \langle 6, 8 \rangle$.

(a) Find the equation of the tangent plane to f at $(4, 3)$.

(b) Use the linear approximation of $f(x, y)$ at $(4, 3)$ to approximate $f(5, 2)$.

(c) What is the rate of change of the function at $(4, 3)$ when moving towards the origin?

(d) Which direction maximizes the rate of change of f at $(4, 3)$?



$$\vec{PQ} = (0, 0) - (4, 3) = -(4, 3) = (-4, -3)$$

$$|\vec{PQ}| = |(-4, -3)| = \sqrt{16 + 9} = \sqrt{25} = 5$$

$$\vec{u} = \frac{1}{|\vec{PQ}|} \vec{PQ} = \frac{1}{5} (-4, -3) = \left(-\frac{4}{5}, -\frac{3}{5}\right)$$

4. Let $f(x, y)$ be a function satisfying $f(4, 3) = 5$ and $\nabla f(4, 3) = \langle 6, 8 \rangle$.

- (a) Find the equation of the tangent plane to f at $(4, 3)$.
- (b) Use the linear approximation of $f(x, y)$ at $(4, 3)$ to approximate $f(5, 2)$.
- (c) What is the rate of change of the function at $(4, 3)$ when moving towards the origin?
- (d) Which direction maximizes the rate of change of f at $(4, 3)$? $\rightarrow u = (6, 8)$

$$\begin{aligned}
 D_u f(4, 3) &= \nabla f(4, 3) \cdot \vec{u} \\
 &= (6, 8) \cdot \left(-\frac{4}{5}, -\frac{3}{5} \right) \\
 &= -\frac{24}{5} - \frac{24}{5} = \boxed{-\frac{48}{5}}
 \end{aligned}$$



5. Let $f(x, y) = \sqrt{x^2 + y^2} \cdot \ln(2x)$.

(a) Find the domain of f .

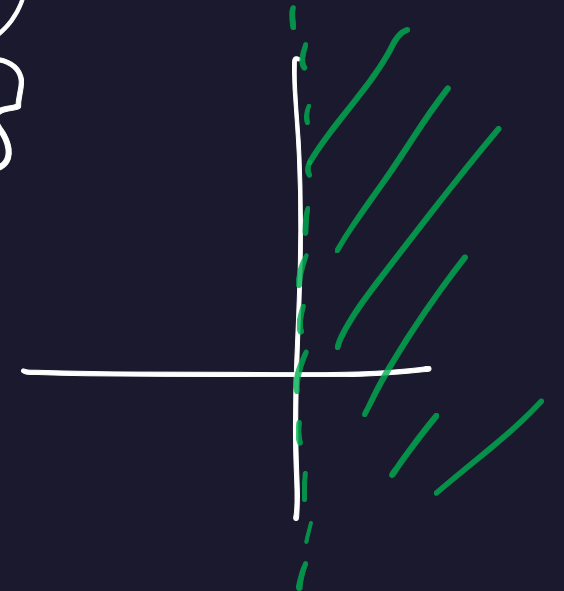
(b) Verify by direct computation that $f_{xy} = f_{yx}$ (also known as *Clairaut's Theorem*).

(a) $x^2 + y^2 \geq 0$ ✓

$\ln(2x) \leadsto 2x > 0 \Rightarrow$

$\left\{ \begin{array}{l} \text{All } (x, y) \\ \text{s.t.} \\ x > 0 \end{array} \right.$

$\text{Domain} = \{ (x, y) : x > 0 \}$



5. Let $f(x, y) = \sqrt{x^2 + y^2} \cdot \ln(2x)$.

(a) Find the domain of f .

(b) Verify by direct computation that $f_{xy} = f_{yx}$ (also known as *Clairaut's Theorem*).

6. Let $f(x, y) = (x^2 - y^2)e^y$ and let $g(t) = \cos(t)$ and $h(t) = \sin(t)$. Use the chain rule to compute the derivative with respect to t of the function $f(g(t), h(t))$.

6. Let $f(x, y) = (x^2 - y^2)e^y$ and let $g(t) = \cos(t)$ and $h(t) = \sin(t)$. Use the chain rule to compute the derivative with respect to t of the function $f(g(t), h(t))$.

7. Let f be a continuous function of two variables which is twice differentiable with the following table of values.

	$f(x, y)$	$f_x(x, y)$	$f_y(x, y)$	$f_{xx}(x, y)$	$f_{xy}(x, y)$	$f_{yy}(x, y)$
$(-1, 2)$	11	0	0	1	5	3
$(1, 4)$	-5	1	0	2	0	4
$(-2, -1)$	6	0	0	-3	0	-1
$(-4, -1)$	0	2	2	1	0	1
$(1, -3)$	2	3	0	-2	5	2

(a) Which points are critical points? **Select ALL that apply.**

(b) Classify each critical point as a local maximum, local minimum or saddle point or explain why there is not enough information to tell.

8. Use the method of *Lagrange Multipliers* to find the maximum and the minimum of $f(x, y) = x^2 + y$ over the ellipse $x^2 + 2y^2 = 8$.

$$\begin{cases} f(x, y) = x^2 + y \\ g(x, y) = x^2 + 2y^2 = 8 \end{cases}$$

$$\nabla f = (2x, 1)$$

$$\nabla g = (2x, 4y)$$

$$2x\lambda - 2x = 0$$

$\downarrow$

$$\begin{cases} \nabla f = \lambda \nabla g \\ g = 8 \end{cases} \quad // \quad \lambda \nabla f = \nabla g$$

$$\Rightarrow \begin{cases} 2x\lambda = 2x \\ \lambda = 4y \\ x^2 + 2y^2 = 8 \end{cases}$$

$$\Rightarrow \begin{cases} 2x(\lambda - 1) = 0 \\ \lambda = 4y \\ x^2 + 2y^2 = 8 \end{cases}$$

CASES

$$\begin{cases} \lambda = 1 \Rightarrow 1 = 4y \text{ or } y = 1/4 \Rightarrow x^2 + 2 \cdot \frac{1}{16} = 8 \Rightarrow x^2 = 8 - \frac{1}{8} = \frac{63}{8} \\ x = 0 \Rightarrow 2y^2 = 8 \Rightarrow y^2 = 4 \Rightarrow y = \pm 2 \end{cases}$$

$$x = \pm \sqrt{63/8}$$

8. Use the method of *Lagrange Multipliers* to find the maximum and the minimum of $f(x, y) = x^2 + y$ over the ellipse $x^2 + 2y^2 = 8$.

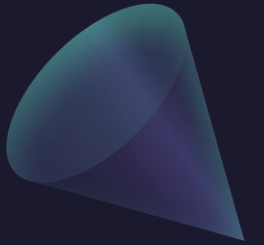
CRITICAL PTS : $\left(\pm\sqrt{\frac{63}{8}}, \frac{1}{4}\right)$ $(0, \pm 2)$ $f = x^2 + y$

$$f\left(\pm\sqrt{\frac{63}{8}}, \frac{1}{4}\right) = \boxed{8.125} \leftarrow \text{MAX}$$

$$f(0, 2) = 2$$

$$f(0, -2) = \boxed{-2} \leftarrow \text{MIN}$$

Questions?



Thank you

Until next time.

